

Draft: the finite excess in transfinite nim multiplication

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Abstract

Conway's field On_2 makes the ordinals below ω^{ω^ω} an algebraic closure of $\mathbb{F}_2$ under nim-arithmetic. Its multiplication is governed by a Kummer tower whose carries are Lenstra's *excess* values: for each odd prime p , the carry is $\alpha_p = \kappa_{f(p)} + m_p$, where the transfinite part $\kappa_{f(p)}$ has a known closed shape (Lenstra, DiMuro) but the finite correction m_p does not. This note consolidates the state of the excess problem (docs/OPEN.md Problem 3 of `ogdoad`): the exact finite-field reformulation of the root test as a multiplicative-order criterion and its structural norm reduction; a complete analysis of the 3-power column via a half-angle splitting in the cyclotomic component fields, certifying $m_r = 1$ for all primes r with $f(r) \in \{3, 9, \dots, 3^6\}$ and forcing $m_p \geq 4$ unconditionally on the $f(p) = 2 \cdot 3^k$ column — now sharpened to $m_p = 4$ *exactly* at every prime current factor tables reach (universally for $k \leq 6$), via a corrected compositum norm $(\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + \omega$ that collapses the $m = 4$ test into the same trinomial field; the empirically exact 0/1/4 candidate rule and the proof that the Lenstra component set alone cannot decide the excess; an exact transverse-norm formulation of every singleton-odd row; two embedded maximal-order problems (the Conway Fermat tower on the even side and explicit Gaussian-period primitivity on the 3-power side), with the latter now reduced to primitivity of the irreducible Dickson iterate $D_{3^k}(X, 1) + 1$ and to an explicit recursive Singer-generator problem in a cubic norm-one torus, proved analytically through the first three prime-order kernel steps and reduced thereafter to an exact character nonvanishing test for a norm-coherent cyclotomic unit; and the fact that proving the candidate would settle Lenstra's open boundedness question, while already requiring those two hard maximal-order inputs. Feasibility of the first open row, $p = 719$, is assessed: a Frobenius-orbit norm recurrence reduces it to arithmetic in $\mathbb{F}_{2^{359}}$, with the cost concentrated in tower-aware Frobenius composition. Each statement carries an explicit claim-level tag.

Claim levels. [**proved**] – unconditional, machine-checked or classically verified; [**certified** $4 \leq k \leq 6$] – proved for the stated range, blocked beyond it only by unfactored cofactors; [**consistent**] – no counterexample found, not proved; [**conjectured**] – explicitly open in the literature or posed in this thread; [**open**] – neither confirmed nor refuted, no feasible attack on record.

1 Notation, implemented state, external data

1.1 Notation

For an odd prime p :

- $f(p) = \text{ord}_p(2)$, the multiplicative order of 2 modulo p ;
- κ_h is the Lenstra/DiMuro tower element indexed by h ;
- $\mathcal{Q}(h)$ is Lenstra's set of prime-power components appearing in κ_h ;

- the *excess* m_p is the least finite m such that $\kappa_{f(p)} + m$ has no p -th root in the relevant finite component field;
- the Kummer carry is $\alpha_p = \kappa_{f(p)} + m_p$.

For the ordinal tower, a row with $\mathcal{Q}(f(p)) = \{q\}$ and finite excess m gives the ordinal sum corresponding to $\kappa_q + m$. Addition of numbers is XOR; $\otimes$ is the nim product.

1.2 Implemented tower state

In the Rust tower (`src/scalar/big/ordinal/tower.rs`), as of 2026-06-13:

- the finite excess rows m_p are now sourced from OEIS A380496 [13], the b-file's 126 known rows (odd primes $3 \leq p \leq 709$); the first 14 rows reproduce DiMuro Table 1 + the formerly locally verified $\alpha_{47} = \omega^{(\omega^7)} + 1$;
- the operational boundary is α_{719} , the first OEIS-unknown row: a carry needing α_{719} or beyond returns **None**;
- the table extends *reach*, not *feasibility*: each α_p is reconstructed in code from $\text{ord}_p(2)$, $\mathcal{Q}(f(p))$, and m_p , but for large p the $\mathcal{Q}$ /finite-subfield reconstruction lives in the degree- e_p component field (e_p in the millions; see the snapshot below), so only the small- e_p rows are materializable end-to-end today.

1.3 External data snapshot (2026-06-09)

- OEIS A380496 [13] has 1417 extended rows: 799 known and 618 unknown; the b-file has 126 initial known rows.
- The first OEIS unknown is row $n = 127$, the 127th odd prime $p = 719$. For $p = 719$: $f(719) = \text{ord}_{719}(2) = 359$ and $\mathcal{Q}(359) = \{359\}$.
- The transfinite-nim-calculator logs [15] record the direct component exponent as $e_{719} = 1,258,230,380$, the practical wall for direct exponentiation.

2 The order criterion and structural norm reduction

2.1 The exact criterion

Throughout, E denotes the degree of the component field containing $\beta = \kappa_{f(p)} + m$, so $f = f(p) \mid E$ and hence $p \mid 2^E - 1$.

Proposition 1 (power criterion). *[proved] $\beta \in \mathbb{F}_{2^E}^\times$ has no p -th root in $\mathbb{F}_{2^E}$ iff $\beta^{(2^E-1)/p} \neq 1$. Hence*

$$m_p = \min\{m \geq 0 : (\kappa_{f(p)} + m)^{(2^E-1)/p} \neq 1\}.$$

Proof. $\mathbb{F}_{2^E}^\times$ is cyclic of order $2^E - 1$ and $p \mid 2^E - 1$; in a cyclic group of order N , β is a p -th power iff $\beta^{N/p} = 1$. $\square$

Corollary 1 (order form). *[proved] If $v_p(2^E - 1) = 1$, then β has no p -th root in $\mathbb{F}_{2^E}$ iff $p \mid \text{ord}(\beta)$, so m_p is the least m with $p \mid \text{ord}(\kappa_{f(p)} + m)$.*

Remark 1 (Wieferich caveat). In general “no p -th root” is equivalent to $v_p(\text{ord } \beta) = v_p(2^E - 1)$, which collapses to $p \mid \text{ord}(\beta)$ only when $v_p(2^E - 1) = 1$. By lifting the exponent,

$$v_p(2^E - 1) = v_p(2^{f(p)} - 1) + v_p(E/f(p)).$$

Thus the hypothesis can fail either at base-2 Wieferich-type primes with $p^2 \mid 2^{f(p)} - 1$ or when $p \mid E/f(p)$. The only known base-2 Wieferich primes are $p = 1093$ and $p = 3511$, both of which lie inside the extended A380496 range. For those rows, and for any row with $p \mid E/f(p)$, the exact test of Proposition 1 (or the full- p -part order condition) must be used. The certifications of Section 3 compute orders exactly, with squarefreeness checks, and are unaffected. The research-note predecessor of this document stated only the order form; the power criterion is the correct general statement.

2.2 Norm reduction

Proposition 2 (norm reduction). *[proved] For $\beta \in \mathbb{F}_{2^E}^\times$ with $f = f(p) \mid E$,*

$$\beta^{(2^E-1)/p} = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta)^{(2^f-1)/p}.$$

Proof. $N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^f}}(\beta) = \beta^{(2^E-1)/(2^f-1)}$, and $((2^E - 1)/(2^f - 1)) \cdot ((2^f - 1)/p) = (2^E - 1)/p$. $\square$

So the $p = 719$ test reduces to: (1) compute $N = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^{359}}}(\kappa_{359} + 1)$ in $\mathbb{F}_{2^{359}}$; (2) test whether $719 \mid \text{ord}(N)$ in $\mathbb{F}_{2^{359}}^\times$, a field of degree 359 in which $719 \mid 2^{359} - 1$.

Proposition 3 (Galois-conjugacy invariance). *[proved] The order of $\kappa_{f(p)} + m$ is constant on its Frobenius orbit in $\mathbb{F}_{2^E}$, and the norm $N = \prod_{i=0}^{E/f-1} \text{Frob}^{if}(\kappa_f + m)$ is determined structurally by the tower equations without materialising the full orbit.*

Proposition 4 (tower pinning). *[proved] The tower equations pin κ_{359} up to Frobenius conjugacy: each composed minimal polynomial along the component chain $359 \rightarrow 179 \rightarrow 89 \rightarrow 11 \rightarrow 5 \rightarrow \text{finite}$ is irreducible (the non- q -th-power property defining the lower excess guarantees this), so κ_{359} is a well-defined conjugacy class and N is a well-defined element of $\mathbb{F}_{2^{359}}$.*

2.3 Exact support and the general singleton-odd obstruction

For $h > 1$, write

$$\Pi_h := \prod_{\substack{\ell \text{ odd prime} \\ \text{ord}_\ell(2)=h}} \ell^{v_\ell(2^h-1)}.$$

This is the full prime-power part of $2^h - 1$ supported at primes whose order of 2 is exactly h ; using the full valuations, rather than only the radical, incorporates the Wieferich cases.

Proposition 5 (exact-order support). *[proved] Let $x = \kappa_h$, let $E = d(x)$, and put*

$$T_h = N_{\mathbb{F}_{2^E}/\mathbb{F}_{2^h}}(x).$$

Then $m_p = 0$ for every odd prime p with $f(p) = h$ if and only if

$$\Pi_h \mid \text{ord}(T_h).$$

For an individual such p , $m_p = 0$ iff $v_p(\text{ord } T_h) = v_p(2^h - 1)$.

Proof. The definition of κ_h gives $h \mid E$. Proposition 2 with $\beta = x$ identifies the p -power test in $\mathbb{F}_{2^E}$ with the same test on T_h in $\mathbb{F}_{2^h}$. In the cyclic group $\mathbb{F}_{2^h}^\times$, failure to be a p -th power is exactly the presence of the full p -primary part of the group order in $\text{ord}(T_h)$. Taking the product over the distinct eligible primes proves the claim. No hypothesis on $p \nmid E/h$ is hidden here: if $c = v_p((2^E - 1)/(2^h - 1))$, the norm exponent gives $v_p(\text{ord } T_h) = \max\{0, v_p(\text{ord } x) - c\}$, so equality with $v_p(2^h - 1)$ is equivalent to $v_p(\text{ord } x) = v_p(2^E - 1)$. $\square$

The singleton-odd case admits a sharper field square. Let r be an odd prime, $q = r^a$, and set $b_r = d(\alpha_r)$, where $\kappa_q^q = \alpha_r$. Since $\alpha_r < \kappa_r$, Lenstra's field description gives $r \nmid b_r$, while the successive Kummer equations have degree r [9, Proposition 1.8]; hence

$$d(\kappa_q) = qb_r, \quad \gcd(q, b_r) = 1.$$

Proposition 6 (singleton transverse-norm criterion). *[proved] Suppose $\mathcal{Q}(h) = \{q\}$ with $q = r^a$ odd. Then $h = qs$ for a divisor $s \mid b_r$. In the field square*

$$F = \mathbb{F}_{2^{qb_r}}, \quad B = \mathbb{F}_{2^{b_r}}, \quad K = \mathbb{F}_{2^{qs}}, \quad C = \mathbb{F}_{2^s},$$

one has $B \cap K = C$ and $F = BK$. Put $x = \kappa_q$ and

$$\Theta_{q,s} := N_{F/K}(x+1) \in K^\times.$$

For every prime p with $f(p) = h$:

(i) *x is a p -th power in F , so $m_p \geq 1$;*

(ii) *$m_p = 1$ iff*

$$v_p(\text{ord } \Theta_{q,s}) = v_p(2^{qs} - 1);$$

(iii) *the easy Kummer norm is*

$$N_{F/B}(x+1) = 1 + \alpha_r,$$

but $p \nmid |B^\times|$, so this norm cannot see the required p -primary obstruction. The norm to K is genuinely transverse.

Proof. Lenstra's component theorem gives $\kappa_h = \sum_{t \in \mathcal{Q}(h)} \kappa_t$ and, for each component t , $t \mid h$ and $\gcd(t, h/t) = 1$ [9, Corollary 2.2]. Thus the singleton assumption identifies $\kappa_h = \kappa_q$. Since $h \mid d(\kappa_h) = d(\kappa_q) = qb_r$, one has $h = qs$ with $s \mid b_r$. The intersection and compositum statements follow from the gcd/lcm rules for finite subfields and $\gcd(q, b_r) = 1$.

Since $p > h \geq r$, one has $p \nmid q$. Also $f(p) = qs \nmid b_r$, so $p \nmid 2^{b_r} - 1$. The relation $x^q = \alpha_r \in B$ therefore gives $p \nmid \text{ord}(x)$, proving (i), including the full p -primary statement. Part (ii) is Proposition 2 applied to $x+1$. Finally, $X^q + \alpha_r$ is the minimal polynomial of x over B , and evaluating it at 1 gives (iii); the invisibility follows again from $p \nmid 2^{b_r} - 1$. $\square$

Thus the singleton-one arm is not an unspecified root search. It is the uniform assertion that the explicitly defined transverse norm $\Theta_{q,s}$ contains the full p -primary part for every prime p of exact order qs . The C_k norms below are the special case $q = 3^k$, $b_r = 2$, $s = 1$; the m_{359} obstruction is the same square at $q = 179$, $b_r = 19580$, $s = 1$. Lenstra's use of Carlitz's primitive-root distribution theorem gives an effective bound at which *some* finite translate works [3, 9]; it does not evaluate this fixed first transverse norm.

2.4 Feasibility for $p = 719$

The component field degree is $E = 2 \cdot 2 \cdot 5 \cdot 11 \cdot 89 \cdot 179 \cdot 359 = 1,258,230,380$; the norm requires $E/f = 3,504,820$ Frobenius steps.

[consistent] The norm can in principle be computed by the binary-splitting Frobenius recurrence $N_{2k} = N_k \cdot \text{Frob}^{f^k}(N_k)$ (iterated-Frobenius technique, von zur Gathen–Shoup [17], Kaltofen–Shoup [8]), requiring about $\log_2(3,504,820) \approx 22$ squarings-with-Frobenius-composition at degree E over $\mathbb{F}_2$. Each Frobenius composition is a modular composition in $\mathbb{F}_2[x]$ modulo a degree- E polynomial; with dense polynomial arithmetic at $E \approx 1.26 \times 10^9$ this is not locally feasible.

[conjectured] (as a cost model, not a theorem) Kummer-tower-aware Frobenius – standard-lattice arithmetic for compatibly embedded finite fields in the style of De Feo–Randriam–Rousseau [5], where q -power Frobenius acts near-monomially per tower level – is the likely 10–100× lever: the sparse factored structure $E = 2 \cdot 2 \cdot 5 \cdot 11 \cdot 89 \cdot 179 \cdot 359$ admits a stepwise Frobenius representation in which each level contributes a small-degree modular composition. A *measured* cost model, not a run, is the correct first deliverable.

[open] Whether the Kummer-tower Frobenius model makes the m_{719} computation feasible in wall-clock hours rather than months on available hardware (a single consumer GPU; GF(2) arithmetic is XOR-sliced and GPU-friendly).

[proved] (2026-06-14 local rehearsal.) The first two dependency rows for the $p = 719$ norm path are locally certified by `experiments/ordinal_excess_probe.py`: $m_{89} = 1$ in the $E = 220$ component field, and $m_{179} = 1$ in the $E = 19,580$ component field via the fixed-base power path (`--deep`, about one minute locally). The remaining local rehearsal row is $m_{359} = 1$, with $E = 3,504,820$. This row is already source-pinned by the imported A380496 table ($359 \leq 709$); what remains open here is an independent certificate on the same code path needed for the genuinely unknown $p = 719$ row.

2.5 The m_{359} rehearsal obstruction

Set

$$d = 19,580, \quad E = 179d = 3,504,820, \quad B = \mathbb{F}_{2^d}, \quad F = \mathbb{F}_{2^E}, \quad L = \mathbb{F}_{2^{179}}.$$

For the $m_{359} = 1$ rehearsal, $f(359) = 179$, $\mathcal{Q}(f(359)) = \mathcal{Q}(179) = \{179\}$, and the candidate element is $\beta = \kappa_{179} + 1$. To represent κ_{179} , the tower uses the lower prime row $f(179) = 178$, $\mathcal{Q}(178) = \{89\}$, $m_{179} = 1$. Thus in the tower model $F = B(x)$ with $x = \kappa_{179}$ and

$$x^{179} = \kappa_{89} + 1 \in B,$$

because the lower row $m_{179} = 1$ has already been certified.

[proved] The cheap Kummer norm down the top tower step is not the 359-test norm. Indeed,

$$N_{F/B}(x + 1) = 1 + (\kappa_{89} + 1) = \kappa_{89}$$

(evaluate the monic polynomial $T^{179} + (\kappa_{89} + 1)$ at $T = 1$, in characteristic 2). But $359 \nmid |B^\times| = 2^{19580} - 1$, since $\text{ord}_{359}(2) = 179$ and $179 \nmid 19580$. Thus the structurally cheap norm lands in exactly the field where the 359-primary obstruction is invisible.

[proved] The norm forced by Proposition 2 is the transverse one

$$N_{F/L}(\beta) = \prod_{i=0}^{19579} \text{Frob}^{179i}(\beta) \in L.$$

Here $\gcd(179, 19580) = 1$, so $B \cap L = \mathbb{F}_2$ and $F = BL$. The Frobenius Frob^{179} preserves B as a set but acts nontrivially on it; it is not the relative Kummer action $x \mapsto \zeta x$ over B . Consequently the known lower certificates for m_{89} and m_{179} do not propagate to m_{359} by taking the easy top-step norm.

[proved] The Wieferich caveat is absent at the next two pressure points: direct modular checks give $2^{179} \not\equiv 1 \pmod{359^2}$ and $2^{359} \not\equiv 1 \pmod{719^2}$. The order form of Corollary 1 is therefore equivalent to the full power criterion for the m_{359} rehearsal and the proposed m_{719} test.

[consistent] (2026-06-16 diagnostic.) Re-running the maintained probe confirmed the quick certificate table and the `--deep` $m_{179} = 1$ certificate. A side norm experiment then computed the correct transverse norm in the current term basis for the feasible analogues: for $p = 89$ the norm to $\mathbb{F}_{2^{11}}$ has support 111/220, and for $p = 179$ the norm to $\mathbb{F}_{2^{178}}$ has support 9691/19580. Thus the target-subfield element is essentially half-dense in this representation; using it as the new fixed base makes the final root-test exponent slower than the direct fixed-base certificate. This is a representation-level diagnostic, not a theorem that no sparse description exists.

[open] The current pure-Python term algebra has no feasible route to a local m_{359} certificate. The direct fixed-base path would have to precompute multiplication by β for $E = 3,504,820$ basis terms and then run an E -bit exponent. The norm path lowers the mathematical problem to a degree-179 target field, but computing that norm still requires the 19580-term transverse Frobenius orbit through $F_{2^{3504820}}$. A practical certificate now appears to require either (i) dense/sliced GF(2) arithmetic such as `gf2x`/NTL, or (ii) a tower-aware Frobenius representation that makes the transverse orbit cheap. This is the obstruction hit in the present pass.

Remark 2 (shared abstraction). The structural primitive needed here – $N_{E/K}(\beta) = \prod_i \text{Frob}^i(\beta)$ computed by a Frobenius-orbit recurrence – is the same primitive that the cyclic-algebra Brauer-invariant bridge (Bridge K, `docs/TBD.md`) requires for reduced norms over a Frobenius-generated cyclic extension. The existing `FieldExtension::relative_norm` does not apply to the term-algebra/transfinite setting, but the shape is identical; factoring out a reusable `relative_norm_over_frobenius_orbit` is an engineering opportunity. This is not a claim that the bounded `Fpn` norm certifies m_{719} .

3 The 3^k family

Write $\zeta := \kappa_{3^k}$ and $h := 3^k$ throughout this section.

3.1 Structure

Proposition 7 (cyclotomic recognition). **[proved]** $\zeta^{3^k} = \kappa_2$, which has order 3; hence ζ is a primitive 3^{k+1} -st root of unity. Since 2 is a primitive root modulo 3^{k+1} , the cyclotomic polynomial $\Phi_{3^{k+1}}(x) = x^{2^h} + x^h + 1$ is irreducible over $\mathbb{F}_2$, the component field is $\mathbb{F}_2(\zeta) = \mathbb{F}_{2^{2h}}$, and it carries the index-2 subfield $L = \mathbb{F}_{2^h}$.

Proposition 8 (half-angle splitting). **[proved]** With $s = (3^{k+1} + 1)/2$ (the inverse of 2 modulo 3^{k+1}),

$$\kappa_{3^k} + 1 = \zeta^s (\zeta^s + \zeta^{-s}),$$

where ζ^s lies in the norm-one circle U (of order exactly 3^{k+1} within the subgroup of order $2^h + 1$) and $(\zeta^s + \zeta^{-s})^2 = \zeta + \zeta^{-1} =: \gamma_k \in L^\times$. Since $\mathbb{F}_{2^{2h}}^\times = L^\times \times U$ with coprime orders,

$$\text{ord}(\kappa_{3^k} + 1) = 3^{k+1} \cdot \text{ord}(\gamma_k), \quad \text{ord}(\gamma_k) \mid 2^{3^k} - 1.$$

Machine-checked corollaries: $(\kappa + 1)^{2^h - 1} = \zeta^{-1}$ and $N_{F/L}(\kappa + 1) = \gamma_k$.

Proposition 9 (translates $m = 2, 3$). **[proved]** $\kappa_{3^k} + 2$ and $\kappa_{3^k} + 3$ split as in Proposition 8 with γ_k replaced by a Galois conjugate. Therefore $\text{ord}(\kappa_{3^k} + m) = 3^{k+1} \cdot \text{ord}(\gamma_k)$ for $m = 1, 2, 3$, and $\text{ord}(\kappa_{3^k}) = 3^{k+1}$.

Proposition 10 (the $2 \cdot 3^k$ exception, unconditionally). **[proved]** Any prime p with $f(p) = 2 \cdot 3^k$ satisfies $p \mid 2^{3^k} + 1$, hence p divides neither 3^{k+1} nor $2^{3^k} - 1$, hence p never divides $\text{ord}(\kappa_{3^k} + m)$ for $m \in \{0, 1, 2, 3\}$. By the order criterion, $m_p \geq 4$. The full primitivity conjecture C_k below is not needed; the splitting alone forces the bound. The argument applies beyond the current tables, e.g. $p = 87211$, $f = 54$.

Proposition 11 (the 3-power column). **[proved]** For a prime r with $f(r) = 3^k$, the values $m \in \{0, 2, 3\}$ are impossible: $m_r = 0$ would require $r \mid \text{ord}(\kappa_{3^k}) = 3^{k+1}$, a power of 3, while $r \equiv 1 \pmod{3^k}$ is a prime distinct from 3; and $m = 2, 3$ share $m = 1$'s order class by Proposition 9. Any failure of C_k therefore gives $m_r \geq 4$ directly.

Proposition 12 (norm tower). **[proved]** $\gamma_{k-1} = \gamma_k^3 + \gamma_k = N_{L_k/L_{k-1}}(\gamma_k)$ (minimal polynomial $X^3 + X + \gamma_{k-1}$), so $\text{ord}(\gamma_{k-1}) \mid \text{ord}(\gamma_k)$. With lifting-the-exponent ($v_r(2^{3^k} - 1) = v_r(2^{f(r)} - 1)$) for old primes $r \neq 3$, full-order parts propagate upward, and conjecture C_k reduces to C_{k-1} plus primitivity at the new primes $r \mid \Phi_{3^k}(2)$.

Proposition 13 (the norm-kernel seam). **[proved]** Put $q = 2^{3^{k-1}}$, so $L_k = \mathbb{F}_{q^3}$ and $L_{k-1} = \mathbb{F}_q$. There is a canonical direct product

$$L_k^\times = L_{k-1}^\times \times U_k, \quad |U_k| = q^2 + q + 1 = \Phi_{3^k}(2),$$

where $U_k = \ker N_{L_k/L_{k-1}}$. If $\gamma_k = a_k b_k$ in this product, then

$$\gamma_{k-1} = N(\gamma_k) = a_k^3,$$

and cubing is an automorphism of $L_{k-1}^\times$. Consequently

$$C_k \iff C_{k-1} \text{ and } b_k \text{ generates } U_k.$$

In particular, the norm recursion determines the old component a_k exactly but supplies no order information about the new component b_k .

Proof. The two displayed factor orders are coprime:

$$\gcd(q - 1, q^2 + q + 1) = \gcd(q - 1, 3) = 1,$$

because $q = 2^{\text{odd}} \equiv 2 \pmod{3}$. The cyclic group $L_k^\times$ therefore splits as asserted. The relative norm is $z \mapsto z^{1+q+q^2}$; it kills U_k and restricts on $L_{k-1}^\times$ to $a \mapsto a^3$, which is bijective because $3 \nmid q - 1$. The final equivalence follows from coprimality of the two factor orders. $\square$

Proposition 14 (primitive Dickson iterate). **[proved]** Let $D_n(X, 1)$ be the Dickson polynomial characterized by $D_n(u + u^{-1}, 1) = u^n + u^{-n}$, and put $f(X) = X^3 + X$. Then

$$P_k(X) := D_{3^k}(X, 1) + 1 = f^{\circ k}(X) + 1$$

is the minimal polynomial of γ_k over $\mathbb{F}_2$. In particular it is irreducible of degree 3^k , and

$$C_k \iff P_k(X) \text{ is a primitive polynomial over } \mathbb{F}_2.$$

The polynomials also obey the coefficient recursion

$$P_{k+1} = P_k^3 + P_k^2 + 1, \quad P_1 = X^3 + X + 1.$$

Proof. The Dickson composition law gives $D_{3^k} = D_3^{\circ k} = f^{\circ k}$ in characteristic two. Since ζ^{3^k} is a primitive cube root of unity,

$$D_{3^k}(\gamma_k, 1) = \zeta^{3^k} + \zeta^{-3^k} = 1.$$

On the other hand, the Frobenius orbit of $\zeta + \zeta^{-1}$ has length 3^k : the least $j > 0$ with $2^j \equiv \pm 1 \pmod{3^{k+1}}$ is 3^k . Thus the displayed monic polynomial, which has degree 3^k , is the minimal polynomial. Its root is primitive precisely when the polynomial is primitive. Finally, writing $D_{3^k} = P_k + 1$ and using $D_3(Y) = Y^3 + Y$ gives $P_{k+1} = (P_k + 1)^3 + (P_k + 1) + 1 = P_k^3 + P_k^2 + 1$. $\square$

Proposition 15 (shifted cyclotomic-unit form). *[proved] Put $x_k = 1 + \zeta \in \mathbb{F}_{2^{2h}}^\times$, where $h = 3^k$ and $N = 3^{k+1}$. Then*

$$x_k^{2^h-1} = \zeta^{-1}, \quad x_k^{2^h+1} = \gamma_k, \quad \text{quad ord}(x_k) = N \text{ ord}(\gamma_k).$$

Consequently

$$C_k \iff \text{ord}(1 + \zeta) = 3^{k+1}(2^{3^k} - 1).$$

Thus the cubic arm is equivalently maximal order for the reduction at 2 of the cyclotomic unit $1 + \zeta_{3^{k+1}}$.

Proof. Because $2^h \equiv -1 \pmod{N}$, $x_k^{2^h} = 1 + \zeta^{-1} = \zeta^{-1}x_k$. Dividing and multiplying by x_k gives the first two identities. The factors $2^h - 1$ and $2^h + 1$ are coprime. In the corresponding direct-product decomposition of $\mathbb{F}_{2^{2h}}^\times$, the first displayed power has exact order N and the second has order $\text{ord}(\gamma_k)$, so the order of x_k is their product. $\square$

Proposition 16 (norm-coherent cyclotomic lift). *[proved] Let $\xi_k = \zeta_{3^{k+1}}$ in characteristic zero and $u_k = 1 + \xi_k$. Then 2 is inert in $\mathbb{Q}(\xi_k)$ and*

$$N_{\mathbb{Q}(\xi_k)/\mathbb{Q}(\xi_{k-1})}(u_k) = u_{k-1}.$$

The norm to the maximal real subfield is

$$N_{\mathbb{Q}(\xi_k)/\mathbb{Q}(\xi_k + \xi_k^{-1})}(u_k) = 2 + \xi_k + \xi_k^{-1},$$

whose reduction modulo the unique prime above 2 is γ_k . Thus Proposition 15 is the residue-field shadow of a norm-coherent cyclotomic-unit system in the cyclotomic $\mathbb{Z}_3$ -tower. Conditional on C_{k-1} , C_k is exactly the assertion that, for every prime $\ell \mid \Phi_{3^k}(2)$, the reduction of u_k is not an ℓ -th power.

Proof. The residue degree of 2 in $\mathbb{Q}(\xi_k)$ is $\text{ord}_{3^{k+1}}(2) = 2 \cdot 3^k = \varphi(3^{k+1})$, proving inertness. The three relative conjugates of ξ_k are $\xi_k, \xi_k\omega$, and $\xi_k\omega^2$, so

$$\prod_{j=0}^2 (1 + \xi_k\omega^j) = 1 + \xi_k^3 = 1 + \xi_{k-1}.$$

Multiplication by the complex-conjugate factor gives the real norm. The final assertion follows from Proposition 15: in a cyclic group an element has the full ℓ -primary part of the group order precisely when it is not an ℓ -th power. $\square$

Proposition 17 (circular-unit and ray-class form). *[proved] Let*

$$F_k = \mathbb{Q}(\xi_k + \xi_k^{-1}), \quad \lambda_k = (1 - \xi_k)(1 - \xi_k^{-1}),$$

let $\mathfrak{p}_k = (2)$ be the unique prime of F_k above 2, and let $\mathcal{C}_k \subset \mathcal{O}_{F_k}^\times$ be the real circular-unit group. If $\sigma(\xi_k) = \xi_k^2$, then

$$c_k := \frac{\sigma(\lambda_k)}{\lambda_k} = (1 + \xi_k)(1 + \xi_k^{-1}) = 2 + \xi_k + \xi_k^{-1}$$

is a circular unit, its Galois conjugates generate a subgroup of 2-power index in $\mathcal{C}_k$, and its reduction modulo $\mathfrak{p}_k$ is γ_k . Since the residue-field unit group has odd order, that 2-power index disappears on reduction. Consequently

$$\overline{\mathcal{C}_k} = \langle \gamma_k \rangle \subset (\mathcal{O}_{F_k}/\mathfrak{p}_k)^\times \simeq \mathbb{F}_{2^{3k}}^\times, \quad (\clubsuit)$$

and hence

$$[\mathbb{F}_{2^{3k}}^\times : \langle \gamma_k \rangle] = [(\mathcal{O}_{F_k}/\mathfrak{p}_k)^\times : \overline{\mathcal{C}_k}]. \quad (\spadesuit)$$

In particular, $\mathcal{C}_k$ is exactly surjectivity modulo 2 of the real circular units of conductor 3^{k+1} .

More precisely, put $E_k = \mathcal{O}_{F_k}^\times$ and let $\text{Cl}_{\mathfrak{p}_k}(F_k)$ be the ray class group of finite modulus $\mathfrak{p}_k$. The ray class exact sequence gives

$$[\mathbb{F}_{2^{3k}}^\times : \langle \gamma_k \rangle] = \frac{|\text{Cl}_{\mathfrak{p}_k}(F_k)|}{|\text{Cl}(F_k)|} [\overline{E_k} : \overline{\mathcal{C}_k}], \quad (\heartsuit)$$

where $[\overline{E_k} : \overline{\mathcal{C}_k}] \mid [E_k : \mathcal{C}_k]$. The circular-unit index theorem identifies the odd part of the latter index with the odd part of $|\text{Cl}(F_k)|$. Thus, for an odd prime $\ell \nmid |\text{Cl}(F_k)|$,

$$v_\ell([\mathbb{F}_{2^{3k}}^\times : \langle \gamma_k \rangle]) = v_\ell\left(\frac{|\text{Cl}_{\mathfrak{p}_k}(F_k)|}{|\text{Cl}(F_k)|}\right). \quad (\diamondsuit)$$

Equation $(\heartsuit)$ names the obstruction but does not reduce its difficulty: the ray class group is defined through this residue-unit quotient, and the class-number exclusion in $(\diamondsuit)$ is itself not known uniformly in k . Finally, with $h = 3^k$, $q = 2^{h/3}$, and $\tau = \sigma^{h/3}$, the relative circular unit

$$\varepsilon_k = \frac{\tau(c_k)}{c_k}, \quad N_{F_k/F_{k-1}}(\varepsilon_k) = 1,$$

reduces to η_k . Thus the exact character obstruction of Proposition 19 is the new relative part of the same ray-class obstruction.

Proof. The class of 2 generates $(\mathbb{Z}/3^{k+1}\mathbb{Z})^\times / \{\pm 1\}$, so σ generates $\text{Gal}(F_k/\mathbb{Q})$. The squares of the standard real circular units are generated, up to -1 , by the quotients $\sigma^j(\lambda_k)/\lambda_k$. Telescoping gives

$$\frac{\sigma^j(\lambda_k)}{\lambda_k} = \prod_{i=0}^{j-1} \sigma^i(c_k),$$

so the conjugates of c_k generate this square subgroup. Squaring is an automorphism of the odd-order residue-field unit group, so the full circular-unit group and this subgroup have the same reduction image. Reduction commutes with σ , which becomes Frobenius at the inert prime 2, and

$$\overline{c_k} = (1 + \zeta)(1 + \zeta^{-1}) = \zeta + \zeta^{-1} = \gamma_k.$$

The reductions of the conjugates are therefore $\gamma_k, \gamma_k^2, \gamma_k^{2^2}, \dots$, and they generate the single multiplicative subgroup $\langle \gamma_k \rangle$. This proves $(\clubsuit)$ and $(\spadesuit)$.

The exact sequence

$$E_k \longrightarrow (\mathcal{O}_{F_k}/\mathfrak{p}_k)^\times \longrightarrow \text{Cl}_{\mathfrak{p}_k}(F_k) \longrightarrow \text{Cl}(F_k) \longrightarrow 1$$

proves $(\heartsuit)$. The divisibility of reduction indices is immediate from $\mathcal{C}_k \subset E_k$, and the circular-unit index theorem gives $(\diamondsuit)$; see Washington [19, Ch. 8]. Finally τ generates $\text{Gal}(F_k/F_{k-1})$, so the norm of its coboundary $\tau(c_k)/c_k$ telescopes to 1. Modulo 2 it is $\gamma_k^q/\gamma_k = \eta_k$. $\square$

Proposition 18 (explicit cubic kernel tower). *[proved] Define $\eta_0 = 0$. For $k \geq 1$ put*

$$q_k = 2^{3^{k-1}}, \quad \eta_k = \gamma_k^{q_k-1}, \quad U_k = \ker \left(N_{\mathbb{F}_{q_k^3}/\mathbb{F}_{q_k}} \right).$$

Then $\eta_k \in U_k$, $|U_k| = q_k^2 + q_k + 1 = \Phi_{3^k}(2)$, and, with $\omega = \zeta^{3^k} \in \mathbb{F}_4$,

$$q_k \equiv 3^k - 1 \pmod{3^{k+1}}, \quad \eta_k = \frac{\omega + \omega^2 \zeta^2}{1 + \zeta^2}, \quad \zeta^2 = \frac{\eta_k + \omega}{\eta_k + \omega^2}.$$

Moreover η_k has the irreducible cubic equation over $\mathbb{F}_{q_k} = \mathbb{F}_{2^{3^{k-1}}}$

$$X^3 + \eta_{k-1}X^2 + (\eta_{k-1} + 1)X + 1 = 0. \quad (*)$$

Equivalently,

$$N(\eta_k) = 1, \quad \text{Tr}(\eta_k) = \eta_{k-1}, \quad e_2(\eta_k) = \eta_{k-1} + 1.$$

It also satisfies

$$\eta_k^{q_k+1} + \eta_k + 1 = 0, \quad \eta_k + 1 = \eta_k^{q_k+1}, \quad \text{ord}(\eta_k + 1) = \text{ord}(\eta_k). \quad (\dagger)$$

Thus η_k lies in the planar Singer difference set

$$\mathcal{H}_k = \{y \in U_k : y^{q_k+1} + y + 1 = 0\}$$

of size $q_k + 1$ inside U_k [12]. Finally,

$$C_k \iff C_{k-1} \text{ and } \text{ord}(\eta_k) = q_k^2 + q_k + 1. \quad (**)$$

(Here C_0 is the tautological assertion that 1 generates $\mathbb{F}_2^\times$.) Thus the new-prime obstruction is the single assertion that the distinguished root of $(*)$ is a generator of the cubic norm-one torus (equivalently, multiplication by it is a Singer cycle).

Proof. The congruence starts with $2 \equiv 3-1 \pmod{9}$ and lifts by cubing: if $a \equiv -1+3^k \pmod{3^{k+1}}$, then $a^3 \equiv -1+3^{k+1} \pmod{3^{k+2}}$. Hence $\zeta^{q_k} = \omega\zeta^{-1}$. Dividing $\gamma_k^{q_k}$ by γ_k now gives the stated Möbius formula; solving it gives the inverse formula.

Put $t = \zeta^2$ and $M(T) = (\omega + \omega^2 T)/(1 + T)$. The same formula one level down says $\eta_{k-1} = M(t^3)$ (also for $k = 1$, when it gives $\eta_0 = 0$). If $a = \eta_{k-1}$ and $X = M(t)$, then

$$\left(\frac{X + \omega}{X + \omega^2} \right)^3 = \frac{a + \omega}{a + \omega^2}.$$

Cross-multiplication and $\omega^2 + \omega + 1 = 0$ reduce exactly to $(*)$. Also $\eta_k^{1+q_k+q_k^2} = \gamma_k^{q_k^3-1} = 1$, so $\eta_k \in U_k$. It cannot lie in $\mathbb{F}_{q_k}$: otherwise its norm-one condition would give $\eta_k^3 = 1$, hence $\eta_k = 1$ because $3 \nmid q_k - 1$; the inverse Möbius formula would then give $t = \omega$, contradicting $\text{ord}(t) = 3^{k+1} > 3$. Thus $(*)$ is irreducible, and its coefficients give the three symmetric identities.

The earlier minimal polynomial $X^3 + X + \gamma_{k-1}$ gives $\text{Tr}_{\mathbb{F}_{q_k^3}/\mathbb{F}_{q_k}}(\gamma_k) = 0$. Dividing $\gamma_k + \gamma_k^{q_k} + \gamma_k^{q_k^2} = 0$ by γ_k yields $1 + \eta_k + \eta_k^{q_k+1} = 0$, proving the first two parts of $(\dagger)$. Since $\gcd(q_k + 1, q_k^2 + q_k + 1) = 1$, exponentiation by $q_k + 1$ is an automorphism of U_k , proving the equality of orders. The cited difference-set identification is the norm-one realization of the planar Singer set.

In the direct product of Proposition 13, raising to $q_k - 1$ kills the old-field component and is an automorphism on U_k , since $\gcd(q_k - 1, q_k^2 + q_k + 1) = 1$. Therefore the kernel component of γ_k generates U_k exactly when η_k does; Proposition 13 gives $(**)$. $\square$

Proposition 19 (no transfer and the exact character test). *[proved] The kernel orders are pairwise coprime: for $1 \leq i < k$,*

$$\gcd(\Phi_{3^i}(2), \Phi_{3^k}(2)) = 1.$$

For each prime $\ell \mid |U_k|$, let $\chi_{k,\ell} : U_k \rightarrow \mu_\ell$ be a character of exact order ℓ . Then

$$\text{Hom}\left(\mathbb{F}_{2^{3k-1}}^\times, \mu_\ell\right) = 1; \quad (\dagger_0)$$

no multiplicative character or power-residue datum from a previous level can transfer an ℓ -part to level k . Moreover,

$$C_k \iff C_{k-1} \text{ and } \chi_{k,\ell}(\eta_k) \neq 1 \text{ for every prime } \ell \mid |U_k|. \quad (\dagger)$$

Equivalently, with

$$\rho_{k,\ell}(A) = \chi_{k,\ell}(A^{q_k-1}) \quad (A \in \mathbb{F}_{q_k}^\times),$$

where $\rho_{k,\ell}$ is trivial on $\mathbb{F}_{q_k}^\times$, the missing assertion is $\rho_{k,\ell}(\gamma_k) \neq 1$ for every such ℓ . This is the smallest power-residue form of the cubic obstruction.

Proof. If a prime r divided two displayed cyclotomic values, then $r \neq 3$ and the standard cyclotomic-divisor criterion would give both $\text{ord}_r(2) = 3^i$ and $\text{ord}_r(2) = 3^k$, a contradiction. Thus every $\ell \mid |U_k|$ is coprime to $|\mathbb{F}_{2^{3k-1}}^\times| = \prod_{j < k} \Phi_{3^j}(2)$, proving $(\dagger_0)$. In the cyclic group U_k , the kernel of an exact order- ℓ character is U_k^ℓ . If $|U_k| = \ell^a m$ with $\ell \nmid m$, then an element has the full ℓ^a in its order exactly when it is not in U_k^ℓ . Applying this for every prime divisor and using $(**)$ proves $(\dagger)$. The formula for $\rho_{k,\ell}$ and $\eta_k = \gamma_k^{q_k-1}$ give the final equivalence. $\square$

Proposition 20 (Capelli form of the surviving lemma). *[proved] Put*

$$A(X) = X^3 + X + 1, \quad B(X) = X^2 + X, \quad R(X) = \frac{A(X)}{B(X)}.$$

Let $Q_0(X) = X$ and, for $k \geq 1$, define the monic binary polynomial

$$Q_k(X) = B(X)^{3^{k-1}} Q_{k-1}(R(X)). \quad (\star)$$

Then Q_k is the minimal polynomial of η_k over $\mathbb{F}_2$ and is irreducible of degree 3^k . Its first four values are

$$\begin{aligned} Q_1(X) &= X^3 + X + 1, \\ Q_2(X) &= X^9 + X + 1, \\ Q_3(X) &= X^{27} + X^{25} + X^{24} + X^{19} + X^{18} + X^{16} + X^{10} + X^9 + X^3 + X + 1, \\ Q_4(X) &= X^{81} + X^{64} + X^{16} + X + 1. \end{aligned}$$

For every prime $\ell \mid \Phi_{3^k}(2)$, the following are equivalent:

- (i) $\chi_{k,\ell}(\eta_k) \neq 1$;
- (ii) η_k is not an ℓ -th power in $\mathbb{F}_{2^{3^k}}^\times$;
- (iii) $X^\ell - \eta_k$ is irreducible over $\mathbb{F}_{2^{3^k}}$;
- (iv) $Q_k(X^\ell)$ is irreducible over $\mathbb{F}_2$.

Thus the surviving character lemma is equivalently the assertion that all the explicitly recursive compositions $Q_k(X^\ell)$, for $\ell \mid \Phi_{3^k}(2)$, are irreducible. At the first composite level $k = 4$, the three targets are the monomial compositions of the displayed pentanomial Q_4 by $\ell \in \{2593, 71119, 97685839\}$.

Proof. Proposition 18 gives $R(\eta_k) = \eta_{k-1}$ and an irreducible cubic for η_k over $\mathbb{F}_{2^{3^{k-1}}}$. Since $\eta_{k-1} \in \mathbb{F}_2(\eta_k)$, induction gives $[\mathbb{F}_2(\eta_k) : \mathbb{F}_2] = 3^k$. Clearing the denominator in the minimal polynomial of η_{k-1} gives $(\star)$, a monic polynomial of degree 3^k having η_k as a root: its leading term is $A(X)^{3^{k-1}} = X^{3^k} + \dots$. It is therefore the minimal polynomial. Direct substitution in $(\star)$ gives the four displayed polynomials.

Here $\Phi_{3^k}(2) = q_k^2 + q_k + 1 \equiv 1 \pmod{3}$, so a prime $\ell \mid \Phi_{3^k}(2)$ is not 3. The cyclotomic-divisor criterion therefore gives $\text{ord}_\ell(2) = 3^k$, and hence $\mu_\ell \subset \mathbb{F}_{2^{3^k}}$. Since ℓ is prime, Kummer theory says that $X^\ell - \eta_k$ either splits off a linear factor or is irreducible, according as η_k is or is not an ℓ -th power. This proves the equivalence of (ii) and (iii), while Capelli's lemma gives the equivalence of (iii) and (iv). The direct product $\mathbb{F}_{2^{3^k}}^\times \simeq \mathbb{F}_{q_k}^\times \times U_k$ and $\ell \nmid (q_k - 1)$ show that being an ℓ -th power in the full field is equivalent, for $\eta_k \in U_k$, to lying in U_k^ℓ . This is the equivalence with (i). $\square$

Remark 3 (Singer membership is not primitivity). The difference-set identification cannot replace the character test. Indeed, exponentiation

$$s_k : U_k \longrightarrow U_k, \quad s_k(y) = y^{q_k+1},$$

is an automorphism of order 6: one has $\gcd(q_k + 1, |U_k|) = 1$ and $(q_k + 1)^3 \equiv -1 \pmod{|U_k|}$, while $(q_k + 1)^2 \equiv q_k \not\equiv 1$. The Singer equation is simply $s_k(y) = y + 1$. Thus its consequence $\text{ord}(y + 1) = \text{ord}(y)$ is automatic order preservation by an automorphism and supplies no order divisor by itself. For example, take $q = 32$ and let y be a root of $X^3 + X^2 + 1$ over $\mathbb{F}_2$. Then y has order 7 and $q + 1 \equiv 5 \pmod{7}$; the defining cubic gives $y^5 + y + 1 = 0$. Hence $y \in \{x : x^{q+1} + x + 1 = 0\}$, but y does not generate the norm-one group of order $q^2 + q + 1 = 7 \cdot 151$. What must be used in the Conway tower is the recursive coefficient η_{k-1} in $(\star)$, not Singer membership alone.

Proposition 21 (prime-step closure). *[proved] If $\Phi_{3^k}(2)$ is prime, then*

$$C_k \iff C_{k-1}.$$

Consequently C_1, C_2, C_3 hold analytically: the three new kernel orders are respectively

$$7, \quad 73, \quad 262657,$$

and all are prime.

Proof. Proposition 18 proves that η_k has degree 3 over $\mathbb{F}_{q_k}$, so in particular $\eta_k \neq 1$. A nonidentity element of a group of prime order generates it, and $(**)$ gives the first claim. The primality of 7 and 73 is immediate. For completeness, $262657 - 1 = 2^9 \cdot 3^3 \cdot 19$, and base 5 gives the Pocklington certificate

$$5^{262656} \equiv 1 \pmod{262657}, \quad \gcd\left(5^{262656/r} - 1, 262657\right) = 1 \quad (r = 2, 3, 19),$$

which proves the third primality assertion. $\square$

Conjecture 1 (C_k : γ -primitivity). γ_k is primitive in $L^\times = \mathbb{F}_{2^{3k}}^\times$.

Proposition 22 (equivalence). *[proved]* For primes r with $f(r) \mid 3^k$ a 3-power:

$$C_k \iff m_r = 1 \text{ for every prime } r \text{ with } f(r) \in \{3, 9, \dots, 3^k\}.$$

The family formula is the 0/1/4 candidate of Section 4 restricted to the 3-power column.

3.2 Certification status

[proved] Proposition 21 proves C_k for $k = 1, 2, 3$ without an order computation.

[certified $4 \leq k \leq 6$ C_k holds for $k = 4, 5, 6$: γ_k is primitive in $\mathbb{F}_{2^{3k}}^\times$, and $\text{ord}(\kappa_{3^k} + 1) = 3^{k+1}(2^{3^k} - 1)$ is verified exactly – product reconstruction, squarefreeness, primality by deterministic Miller–Rabin below 3.3×10^{24} and MR-64 probable-primality above, an independent sieve re-deriving the small factors, cross-validation against the independent term algebra for $k = 1, 2$ and against the calculator rows $m_{2593} = 1$, $m_{487} = 1$ (`experiments/cyclotomic_3k_family.py`, `experiments/excess/`).

[consistent] $k = 7, 8$: every known prime factor of $\Phi_{3^7}(2)$ and $\Phi_{3^8}(2)$ divides $\text{ord}(\gamma_k)$. Full certification is blocked by the unfactored cofactors (FactorDB status composite-no-factor), including their unknown prime multiplicities; no failure has been found.

[open] Whether ECM or GNFS can factor the $\Phi_{3^7}(2)$ and $\Phi_{3^8}(2)$ cofactors within a realistic budget, converting $k = 7, 8$ from CONSISTENT to CERTIFIED.

3.3 Certified excess rows produced

The following $m_r = 1$ rows are certified by the order criterion independently of the cofactor issue (analysis-level results; they are not new Rust carries):

f	primes r with $m_r = 1$
27	262657
81	71119, 97685839
243	16753783618801, 192971705688577, 3712990163251158343
729	80191, 97687, 379081, P42, P90
2187	39367, 7606246033, 263196614521, 529063556041
6561	209953, 1299079, 70063267397606709277393

(P42, P90 denote the certified/probable primes of 42 and 90 decimal digits dividing $\Phi_{3^6}(2)$.)

4 The 0/1/4 candidate

Conjecture 2 (the 0/1/4 rule). [**consistent**] (not proved)

$$m_p = \begin{cases} 0 & \text{if } \mathcal{Q}(f(p)) \text{ is not a singleton odd prime-power,} \\ 4 & \text{if } f(p) = 2 \cdot 3^k, k \geq 1, \\ 1 & \text{otherwise (singleton odd prime-power } \mathcal{Q}(f(p))). \end{cases}$$

The rule matches all 950 calculator records with known $\mathcal{Q}$ -sets and every OEIS-known row covered by those $\mathcal{Q}$ -sets.

4.1 Two maximal-order problems already contained in the rule

The candidate contains strong explicit maximal-order assertions even before the non-cyclotomic singleton chains are reached. This prevents a proof from following from field degree or irreducibility alone.

Proposition 23 (the singleton-even arm is Conway–Fermat maximality). [**proved**] For $n \geq 0$, put

$$h_n = 2^{n+1}, \quad c_n = \kappa_{h_n} = 2^{2^n}, \quad E_n = \mathbb{F}_{2^{h_n}}, \quad F_n = 2^{2^n} + 1.$$

With $E_{-1} = \mathbb{F}_2$, let

$$\delta_n = \text{ord}(c_n E_{n-1}^\times) \quad \text{in} \quad E_n^\times / E_{n-1}^\times.$$

Then $1 < \delta_n \mid F_n$, and the $m = 0$ arm of Conjecture 2 for every prime $p \mid F_n$ is equivalent to

$$\delta_n = F_n.$$

Proof. The Conway finite-nimber tower satisfies

$$c_n^2 + c_n = c_{n-1} c_{n-2} \cdots c_0$$

(with the empty product 1). Thus E_n/E_{n-1} is quadratic and $c_n \notin E_{n-1}$, so the coset order is nontrivial. Its ambient quotient has order

$$\frac{2^{2^{n+1}} - 1}{2^{2^n} - 1} = F_n,$$

which proves $1 < \delta_n \mid F_n$; equivalently the relative norm is $c_n^{F_n} = c_{n-1} \cdots c_0$.

If $p \mid F_n$, then $2^{2^n} \equiv -1 \pmod{p}$, hence $f(p) = 2^{n+1} = h_n$. Fermat numbers are pairwise coprime, so the entire p -primary part of $|E_n^\times|$ lies in the quotient by $E_{n-1}^\times$. The full power criterion therefore gives $m_p = 0$ iff $v_p(\delta_n) = v_p(F_n)$. Requiring this for every prime divisor of F_n is exactly $\delta_n = F_n$. $\square$

When F_n is prime, irreducibility and $\delta_n > 1$ force $\delta_n = F_n$, recovering Lenstra’s Fermat-prime rule. When F_n is composite, irreducibility forces only one nontrivial divisor: it does not force every prime-power factor. The condition $\delta_n = F_n$ is exactly the maximal-coset-order hypothesis studied for Conway’s quadratic tower by Popovych [16]. It is strictly weaker than primitivity of c_n , which already fails at $n = 2$ since $\text{ord}(c_2) = 5 \cdot 17 = 85 < 255$; Popovych uses the all-level hypothesis instead to prove that products such as $c_0 c_n$ are primitive. Wiedemann’s related iterated quadratic tower poses the analogous full-Fermat-order problem [18]. Thus the apparently easy “singleton even implies zero” clause already contains a longstanding explicit maximal-coset-order problem.

Proposition 24 (the 3-power arm is Gaussian-period primitivity). *[proved] Fix $k \geq 1$. The $m = 1$ arm of Conjecture 2 for every prime p with $f(p) \in \{3, 9, \dots, 3^k\}$ is equivalent to C_k : the Gaussian period*

$$\gamma_k = \zeta + \zeta^{-1} \in \mathbb{F}_{2^{3^k}}^\times, \quad \zeta^{3^{k+1}} = 1,$$

is primitive.

Proof. Propositions 8 and 12 show that the old prime-power factors of $2^{3^k} - 1$ propagate from lower levels. For a prime p with $f(p) = 3^j$, the full power criterion says $m_p = 1$ iff $v_p(\text{ord } \gamma_j) = v_p(2^{3^j} - 1)$. Combining these statements for all $j \leq k$ gives every prime-power factor of $2^{3^k} - 1$, which is equivalent to primitivity; the converse is immediate. $\square$

The two arms are instances of the same projective statement. The quadratic arm asks that the class of c_n generate $E_n^\times / E_{n-1}^\times$. In the cubic arm the natural map

$$U_k \longrightarrow \mathbb{F}_{q_k}^\times / \mathbb{F}_{q_k}^\times$$

is an isomorphism, and the increment from C_{k-1} to C_k asks that the class of γ_k generate this quotient. Equivalently, the projective class of the multiplication-by- γ_k matrix generates a Singer cycle in $\text{PGL}_3(q_k)$. The 0/1/4 rule therefore contains a quadratic and a cubic maximal-projective-order conjecture, rather than two unrelated order coincidences.

Proposition 25 (cyclotomic-type residue reduction). *[proved] Let $h = 3^k$ and let ℓ be any prime divisor of $2^h - 1$. Then*

$$\gamma_k = \zeta^{-1}(1 + \zeta)^2$$

is an ℓ -th power in $\mathbb{F}_{2^h}$ if and only if $1 + \zeta$ is an ℓ -th power in $\mathbb{F}_{2^{2h}}$. The same equivalence holds at every power $\ell^a \mid 2^h - 1$; however, non- ℓ -power already implies that γ_k has the full ℓ -primary part of $2^h - 1$ in its order.

Proof. Because $\ell \mid 2^h - 1$, one has $\ell \nmid 2^h + 1$. Extension from $\mathbb{F}_{2^h}^\times$ to $\mathbb{F}_{2^{2h}}^\times$ therefore preserves the ℓ -primary quotient. Also $\ell \neq 3$, so exponentiation by ℓ is an automorphism of the 3^{k+1} -group generated by ζ ; hence ζ^{-1} is an ℓ -th power. Finally, squaring is an automorphism of every ℓ -primary quotient for odd ℓ . The displayed factorization then proves both assertions. $\square$

Even in the standard prime-conductor case, general results on multiplicative orders of Gaussian periods provide large lower bounds but not maximal order [1]. For the present prime-power conductor, the norm recursion $\gamma_k^3 + \gamma_k = \gamma_{k-1}$ propagates the old factors only. At level k , a proof must still show that every new prime-power factor of $\Phi_{3^k}(2)$ divides $\text{ord}(\gamma_k)$. That is the precise new-prime obstruction; degree 3^k guarantees at most that some exact-order factor occurs, not that all of them do.

The Dickson formulation cleanly separates what the existing theory does and does not supply. Mullin–Mahalanobis give the corresponding irreducibility criterion for $D_{3^k}(X, 1) + 1$ [11]; irreducibility is not primitivity. For prime conductor, Gao–Vanstone’s analogous Gaussian-period primitivity statement remains conjectural [7], while Breuer shows that small divisors of the period index can reflect units and class numbers in an intermediate real quadratic field [2]. The real 3-power cyclotomic tower here has only odd-degree intermediate fields, so that quadratic norm argument does not apply. Proposition 18 supplies the appropriate cubic replacement and places the root on a Singer difference set, but neither the three symmetric coefficients nor the difference-set property proves that this distinguished member generates U_k ; the explicit $q = 32$ example in Remark 3 rules out that shortcut.

4.2 The component set alone is insufficient

[proved] $\mathcal{Q}(f(p))$ alone does not determine m_p :

$\mathcal{Q}$	$m = 4$	$m = 1$
$\{9\}$	$p = 19, f = 18$	$p = 73, f = 9$
$\{81\}$	$p = 163, f = 162$	$p = 2593, f = 81$
$\{243\}$	$p = 1459, f = 486$	$p = 487, f = 243$

[proved] The split for $\mathcal{Q} = \{9\}$ is explained by the order criterion: $\text{ord}(\kappa_9 + 1) = 3^3 \cdot (2^9 - 1) = 27 \cdot 511 = 13797$ and $511 = 7 \cdot 73$, so $73 \mid \text{ord}(\kappa_9 + 1)$ but $19 \nmid \text{ord}(\kappa_9 + 1)$; adding $m = 4$ changes the order and introduces the factor 19.

[proved] The lower bound of the exception column is unconditional: $m_p \geq 4$ for all p with $f(p) = 2 \cdot 3^k$ (Proposition 10).

[conjectured] The matching upper bound $m_p \leq 4$ on that column; equivalent to C_k for the primes in the exception column.

4.3 The $m = 4$ upper bound: settled at every visible prime

The $m = 4$ translate $\kappa_{3^k} + 4$ involves the number $4 \in \mathbb{F}_{16}$, which does not lie in $\mathbb{F}_2(\zeta)$ (as $4 \nmid 2 \cdot 3^k$); the translate lives in the degree- $4 \cdot 3^k$ compositum $\mathbb{F}_{2^{4h}}$, $h = 3^k$. (June 2026 pass; probe `experiments/exception_column_m4.py`.)

Proposition 26 (the corrected norm). [proved] Let $\sigma = \text{Frob}^{2h}$ generate $\text{Gal}(\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}})$. Then $\sigma(4) = 5$, and with $\omega := \zeta^h = \kappa_2$,

$$N := N_{\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}}}(\kappa_{3^k} + 4) = (\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + \omega.$$

Proof. $\mathbb{F}_{16} \cap \mathbb{F}_{2^{2h}} = \mathbb{F}_4$ because $\gcd(4, 2 \cdot 3^k) = 2$, so $\sigma|_{\mathbb{F}_{16}}$ is the nontrivial element of $\text{Gal}(\mathbb{F}_{16}/\mathbb{F}_4)$, which swaps the two roots 4, 5 of the Artin–Schreier minimal polynomial $y^2 + y + \omega$ of 4 over $\mathbb{F}_4$. (Equivalently: $2h \equiv 2 \pmod{4}$ and the Frobenius orbit of 4 is $4 \rightarrow 6 \rightarrow 5 \rightarrow 7$, so $\sigma(4) = 4^{2^2} = 5$.) Then $(\kappa + 4)(\kappa + 5) = \kappa^2 + \kappa + (4 \otimes 5)$ and $4 \otimes 5 = 4^2 + 4 = \omega$ by the same Artin–Schreier relation. $\square$

Remark 4 (erratum). The 2026-06-10 draft of this note stated the norm as $(\kappa + 4)(\kappa + \text{Frob}^2(4)) = (\kappa + 4)(\kappa + 6)$. The conjugate $6 = 4^2$ is $\text{Frob}^1(4)$, one squaring short of $\text{Frob}^2(4) = 5$. The slip is visible without computation: $4 \otimes 6 = 14 \notin \mathbb{F}_4$, so $(\kappa + 4)(\kappa + 6) \notin \mathbb{F}_{2^{2h}}$ and cannot be a norm.

The corrected norm collapses the $m = 4$ test into the *same* sparse trinomial model $\mathbb{F}_2[x]/(x^{2h} + x^h + 1)$ as the rest of the 3-power analysis: no compositum arithmetic is needed.

Corollary 2 (in-field criterion). [proved] For a prime p with $f(p) = 2 \cdot 3^k$ and $v_p(2^{2h} - 1) = 1$, write $\overline{N} = N^{2^h}$ and $M_k := \overline{N}/N = N^{2^h-1} \in U$. Then

$$m_p = 4 \iff p \mid \text{ord}(M_k) \iff M_k^{(2^h+1)/p} \neq 1.$$

Proof. $m_p \geq 4$ is Proposition 10, so $m_p = 4$ iff $\kappa + 4$ has no p -th root. By Propositions 1 and 2 with $E = 4h$, $f = 2h$ (and $v_p(2^{4h} - 1) = v_p(2^{2h} - 1) = 1$ by lifting the exponent, since $p \nmid 2$), that holds iff $N^{(2^{2h}-1)/p} \neq 1$. Since $\mathbb{F}_{2^{2h}}^\times = L^\times \times U$ with coprime orders, $p \mid 2^h + 1$, and $(2^{2h} - 1)/p = (2^h - 1) \cdot ((2^h + 1)/p)$, the test equals $M_k^{(2^h+1)/p} \neq 1$. $\square$

Conjecture 3 (D_k : the column analogue of C_k). *The prime-to-3 part of $\text{ord}(M_k)$ is full: $((2^h + 1)/3^{k+1}) \mid \text{ord}(M_k)$. Equivalently (given the per-prime squarefreeness audits): $m_p = 4$ for every prime p with $f(p) = 2 \cdot 3^k$.*

[**certified** $k \leq 6$] D_k holds for $k = 2, \dots, 6$: every prime factor of the fully factored $\Phi_{2 \cdot 3^k}(2)$ divides $\text{ord}(M_k)$. Hence $m_p = 4$ exactly — not merely ≥ 4 — for every prime p with $f(p) \in \{18, 54, 162, 486, 1458\}$:

f	primes p with $m_p = 4$
18	19
54	87211
162	163, 135433, 272010961
486	1459, 139483, 10429407431911334611, 918125051602568899753
1458	227862073, 3110690934667, 216892513252489863991753, 1102099161075964924744009, P78

(P78 is the certified 78-digit prime of $\Phi_{2 \cdot 3^6}(2)$; PRP-local here, factordb marks it proven.) The anchor rows $m_{19} = m_{163} = m_{1459} = 4$ (DiMuro / calculator) are reproduced, never assumed; the remaining eleven rows are new. Independent cross-checks: the term algebra of `experiments/ordinal_excess_probe.py` re-derives the $p = 19$ root pattern ($m = 0, \dots, 3$ root, $m = 4$ no root) and $m_{87211} = 4$; an explicit Artin–Schreier compositum model $\mathbb{F}[y]/(y^2 + y + \omega)$ verifies $\sigma(4) = 5$, the norm identity, and the full direct power test $(\kappa + 4)^{(2^{4h}-1)/p} \neq 1$ on the smallest prime of each level $k \leq 6$.

[**consistent**] $k = 7, 8$: every *known* prime factor of $\Phi_{2 \cdot 3^7}(2)$ and $\Phi_{2 \cdot 3^8}(2)$ — seven and five primes respectively; factordb CF entries verified locally by divisibility, primality, and order tests, the small ones re-derived by sieve, and the list cross-audited against the raw factordb API after an adversarial review caught a dropped 43-digit $k = 7$ prime — also has $m_p = 4$. An $m_p \geq 5$ example, if one exists, now hides strictly inside the unfactored cofactors; their prime multiplicities are likewise unknown until factorization.

Proposition 27 (twisted norm tower). [**proved**] *With $\eta = \zeta^3 = \kappa_{3^{k-1}}$,*

$$N_{\mathbb{F}_{2^{2 \cdot 3^k}}/\mathbb{F}_{2^{2 \cdot 3^{k-1}}}}(N_k) = \eta^2 + \omega^2 \eta + 1,$$

which is neither $N_{k-1} = \eta^2 + \eta + \omega$, nor one of its Frobenius conjugates (their exponent pattern is $(2^{j+1}, 2^j)$), nor a scaling $N_{k-1}(\lambda\eta)$ with $\lambda \in \mu_3$.

Proof. The generator τ of the degree-3 step sends $\zeta \mapsto \zeta^c$ with $c = 2^{2 \cdot 3^{k-1}}$ a primitive cube root of unity modulo 3^{k+1} ; the cube roots are $1 + a3^k$, so $\tau(\zeta) = \omega^a \zeta$ ($a \neq 0$) while $\tau(\omega) = \omega$. Hence the norm is $\prod_{r^3=\eta} (r^2 + r + \omega)$ over the three cube roots r of η . Writing α, β for the two roots of $T^2 + T + \omega$ (so $\alpha + \beta = 1$, $\alpha\beta = \omega$), each factor is $(r + \alpha)(r + \beta)$, and the product collapses by the resolvent

$$\prod_{r^3=\eta} (r^2 + r + \omega) = (\alpha^3 + \eta)(\beta^3 + \eta) = \eta^2 + (\alpha^3 + \beta^3)\eta + (\alpha\beta)^3 = \eta^2 + \omega^2 \eta + 1,$$

since $\alpha^3 + \beta^3 = (\alpha + \beta)^3 + \alpha\beta(\alpha + \beta) = 1 + \omega = \omega^2$ and $(\alpha\beta)^3 = \omega^3 = 1$. Machine-checked at $k = 2, 3, 4$. $\square$

So, unlike the γ tower of Proposition 12, old-prime full-order parts do *not propagate by that argument*, and D_k is genuinely per-level for the present method: the certification at each k stands on its own. Structurally, at the degree-3 step with $Q = 2^{2 \cdot 3^{k-1}}$, the relative norm kernel has order

$$Q^2 + Q + 1 = \Phi_{3^k}(2)\Phi_{2 \cdot 3^k}(2).$$

Its gcd with $Q - 1$ is 3, so there is no direct-product splitting at the 3-part as in Proposition 13; nevertheless, for every $\ell \neq 3$ the new ℓ -Sylow subgroups of both columns lie entirely in the norm kernel. This is the prime-to-3 analogue of Proposition 13 and explains why a lower-level norm identity cannot determine D_k .

Remark 5 (the 3-part). **[consistent]** The 3-part of $\text{ord}(M_k)$ carries no column information but is strikingly regular in range: $v_3(\text{ord } M_k) = k + 1$ (full) for odd k and $k - 1$ (deficiency exactly 3^2) for even k , for $2 \leq k \leq 6$. No explanation is on record.

5 Boundedness

[conjectured] (Lenstra [9]) m_p is globally bounded. Lenstra gave unconditional lower-bound rules (singleton odd Q forces positive excess; $f(p) = 2 \cdot 3^k$ forces excess at least 4) but left absolute boundedness open.

[proved] If the 0/1/4 candidate holds, then $m_p \leq 4$ for all p ; proving Conjecture 2 would settle boundedness.

[proved] The 3-power column cannot produce $m \in \{2, 3\}$ (Proposition 11); any C_k failure jumps directly to $m_r \geq 4$.

[proved] (conditional on C_k) For every prime r with $f(r) = 3^j$, $j \leq k$: C_k implies $m_r = 1$. With C_k certified for $k \leq 6$, all primes in the 3-power column with $f(r) \leq 6561$ have $m_r = 1$ unconditionally.

[certified $k \leq 6$] (June 2026 pass) On the $2 \cdot 3^k$ exception column the matching upper bound is now certified at every visible prime: $m_p = 4$ exactly for every prime with $f(p) = 2 \cdot 3^k$, $k \leq 6$, and for every known prime at $k = 7, 8$ (Section 4.3). Boundedness on the exception column therefore holds unconditionally through $f = 1458$ and is equivalent to D_k (Conjecture 3) beyond.

[open] Boundedness outside the 3-power and $2 \cdot 3^k$ columns. The non-cyclotomic singleton chains (e.g. the 11-chain and the 23, 29, 47 components) have $m = 1$ rows in the calculator data and OEIS, but no order-criterion proof covers them the way C_k covers the 3-power column.

[open] Whether any prime p has $m_p \geq 5$. No counterexample is known; the question is not known to be decidable by any current structural argument outside the 3-adic columns.

[proved] A proof of the full 0/1/4 candidate would in particular prove both the Conway–Fermat maximality statement $\delta_n = F_n$ at every quadratic level (Proposition 23) and the Gaussian-period primitivity statement C_k at every cubic level (Proposition 24). These are independent new-factor problems; neither follows from the existing norm recursions.

6 Ranked next moves

1. *Conway–Fermat quotient lemma (highest analytical priority)*. Prove or refute $\delta_n = F_n$ in Proposition 23. The norm identity gives only $1 < \delta_n \mid F_n$; the missing step is to exclude each proper prime-power quotient of F_n . A proof must use more than irreducibility, and would settle the entire singleton-even arm.

2. *New-prime lemma for the 3-power spine.* For every $\ell^a \mid \Phi_{3^k}(2)$, prove $v_\ell(\text{ord } \gamma_k) = a$. The norm tower already carries all old factors, so this single per-level statement is exactly C_k . Equivalently, prove that the distinguished root η_k of $X^3 + \eta_{k-1}X^2 + (\eta_{k-1} + 1)X + 1$ generates the norm-one torus U_k . The sharp analytical languages are therefore either the ℓ -power residue symbol of the cyclotomic unit $1 + \zeta$ or the character theory of this explicit cubic trace/norm fibre, not another order census.
3. *General transverse reciprocity.* Evaluate the power-residue symbol of $\Theta_{q,s}$ from Proposition 6. A theorem forcing its full exact-order support would prove the singleton-one arm simultaneously for the 11-chain and the 23, 29, 47, 179, ... components. The easy Kummer norm is provably the wrong object.
4. *Exception-column new-prime lemma.* Prove D_k for the prime-to-3 part of $\text{ord}(M_k)$. The twisted norm formula in Proposition 27 rules out direct induction on k ; a residue-symbol calculation for $N_k = \zeta^2 + \zeta + \omega$ must account for the twist.
5. *Numerical work only as falsification and audit.* The m_{359}/m_{719} transverse norm, complete factorizations at $k = 7, 8$, and exact-order censuses remain useful counterexample searches and checks of any proposed lemma, but they no longer lead the proof program.

7 Provenance and validation

7.1 Independent oracle

`experiments/ordinal_excess_probe.py` is a small local term-algebra oracle – not a replacement for CGSuite [14] or the C++ calculator [15], but a way to verify the first subtle cases without using the Rust production tower as its own oracle. It implements the impartial term algebra used by the calculators, a multiplicative-order test for small component fields, and a fixed-base exponentiation path (ported from the calculator’s strategy) for targeted root tests where full order factorization is unnecessary. Current probe output:

p	m	$\mathcal{Q}$	root?	
7	0	{3}	yes	
7	1	{3}	no	
19	1	{9}	yes	
19	4	{9}	no	
73	1	{9}	no	
89	1	{11}	no	
47	1	{23}	no	
179	1	{89}	no	(--deep)

The $p = 47$ line certifies $m_{47} = 1$ using only lower verified rows; since $f(47) = 23$ and $\mathcal{Q}(23) = \{23\}$, the carry is $\alpha_{47} = \kappa_{23} + 1 = \omega^{(\omega^7)} + 1$, the value now implemented in `tower.rs`. The $p = 179$ line is the first locally certified dependency row beyond the quick probe path; it still uses only the already-certified lower row $m_{89} = 1$.

7.2 The α_{47} promotion

Superseded 2026-06-13: this pass moved the boundary to α_{53} by promoting a single locally verified row; the finite excess table is now the full OEIS A380496 b-file (126 rows, odd primes $3 \leq p \leq 709$), so the refusal boundary sits at α_{719} and the boundary test is `boundary_returns_none_past_prime_709`. The original pass is recorded below for provenance.

The shipped state at the time: `src/scalar/big/ordinal/tower.rs` carries α_{47} with the landmark test `locally_verified_alpha_47_landmark` and moves the refusal boundary to α_{53} ; the documented boundaries in `nim.rs`, `mod.rs`, `src/games/nimber_game.rs`, `README.md`, and `docs/OPEN.md` were updated in step. The full gate ran clean: `cargo fmt --check`, `cargo test`, `cargo check/clippy` with and without the `python` feature (warnings denied), and the probe under `py_compile`.

7.3 Supporting probes

The 3^k -family certification chain is `experiments/cyclotomic_3k_family.py` (committed, maintained), joined 2026-06-12 by `experiments/exception_column_m4.py` (the $2 \cdot 3^k$ exception column, Section 4.3; committed, maintained, stdlib-only, ~ 2 minutes), with the research-run probes under `experiments/excess/` (machine-generated, rescued from ephemeral storage; triage before citing). The analysis in Sections 2–5 synthesizes the 2026-06-10 parallel research run plus the 2026-06-12 exception-column pass; each claim above carries its own tag, and no claim depends on an unverified solver.

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